

## **Week 3 Report**

### **PASSWORD CRACKING REPORT**

#### **JTR & NETWORKWALKS PASSWORD CRACKING TOOLS**

#### **Project information**

- Name: Rachel Debbarma
- Date: 27 August 2026
- Modules completed:
  - W3-PM1 — Password Cracking with JTR
  - W3-PM2 — Password Cracking with Networkwalks Tools
- Target: Locked PDF
- Authorization: Educational lab / supplied practice file

#### **1. Introduction**

This report covers the password-cracking activities completed during Week 3 of my Cybersecurity & Ethical Hacking internship at Networkwalks. Two project modules were completed using different approaches to recover the password of the supplied protected PDFs. The first module focused on John the Ripper (JTR) and Johnny, while the second module used the Networkwalks Hash Calculator and Password Cracker.

Both modules demonstrated the process of extracting a password hash, using password-cracking techniques to identify the original password, and verifying the result by opening the protected PDF. These activities helped me understand how password-cracking tools work and why weak and predictable passwords can create security risks.

#### **2. Tools Used**

| <b>Tool</b>                          | <b>Purpose</b>                            |
|--------------------------------------|-------------------------------------------|
| <b>John the Ripper</b>               | Password recovery/testing                 |
| <b>Johnny</b>                        | GUI for John the Ripper                   |
| <b>PDF Hash Extraction Tool</b>      | Extract PDF password hash                 |
| <b>Networkwalks Hash Calculator</b>  | Extract \$pdf\$ hash from protected PDF   |
| <b>Networkwalks Password Cracker</b> | Perform dictionary-based password testing |
| <b>Adobe Acrobat Reader</b>          | Verify recovered password                 |

#### **3. Activities Performed**

##### **3.1 Password Cracking with JTR**

- JTR installation
- Johnny configuration

- PDF hash acquisition
- Loading the hash
- Starting the attack
- Password found
- Opening the PDF

### 3.2 Password Cracking with Networkwalks Tools

- Upload PDFs
- Generate \$pdf\$... hash
- Copy complete hash
- Open Password Cracker
- Paste hash
- Start attack
- Password found
- Open PDF

## 4. Risk Analysis / Impact

| # | Risk / Finding                   | Evidence / Observation                                       | Potential Impact                                         | Risk   |
|---|----------------------------------|--------------------------------------------------------------|----------------------------------------------------------|--------|
| 1 | Weak PDF password                | password1 was recovered successfully                         | Common passwords can be recovered quickly                | Medium |
| 2 | Dictionary attack effectiveness  | Password was found using dictionary-based testing            | Weak passwords are vulnerable to common-password attacks | Medium |
| 3 | Password-protected file exposure | PDF password could be recovered from its hash representation | Unauthorized access may occur if the password is weak    | Medium |

These are **lab observations**, not vulnerabilities in Networkwalks' systems.

## 5. Recommendations

Things like:

- Use long and unique passwords.
- Avoid common passwords such as password1.
- Avoid reusing passwords.
- Use a password manager.
- Use stronger password policies for sensitive files.
- Review protection mechanisms for sensitive documents.
- Perform password-strength testing only with authorization.

## 6. Conclusion

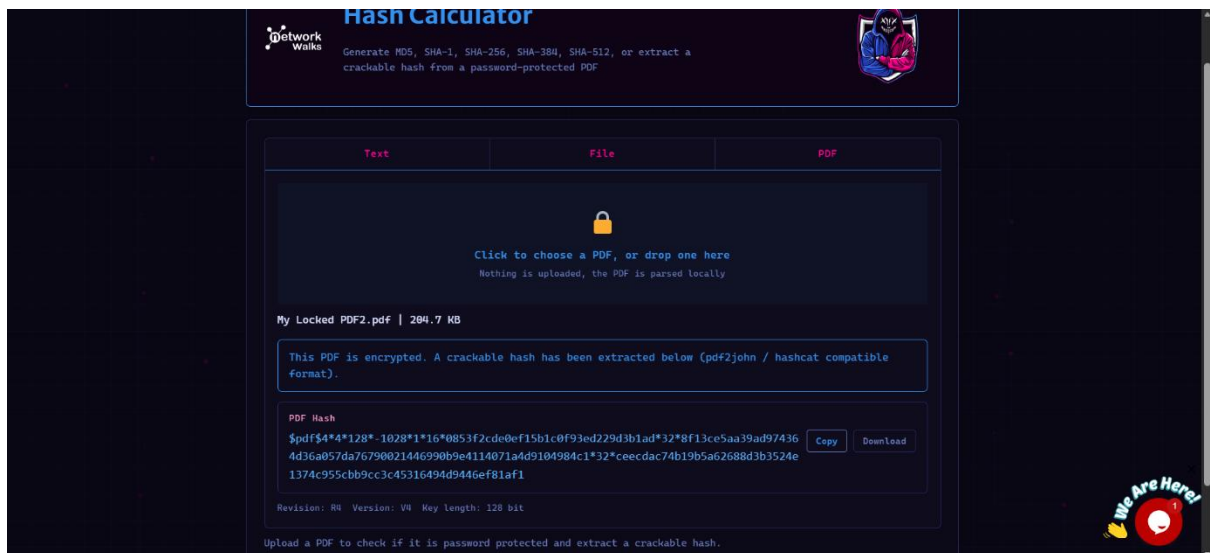
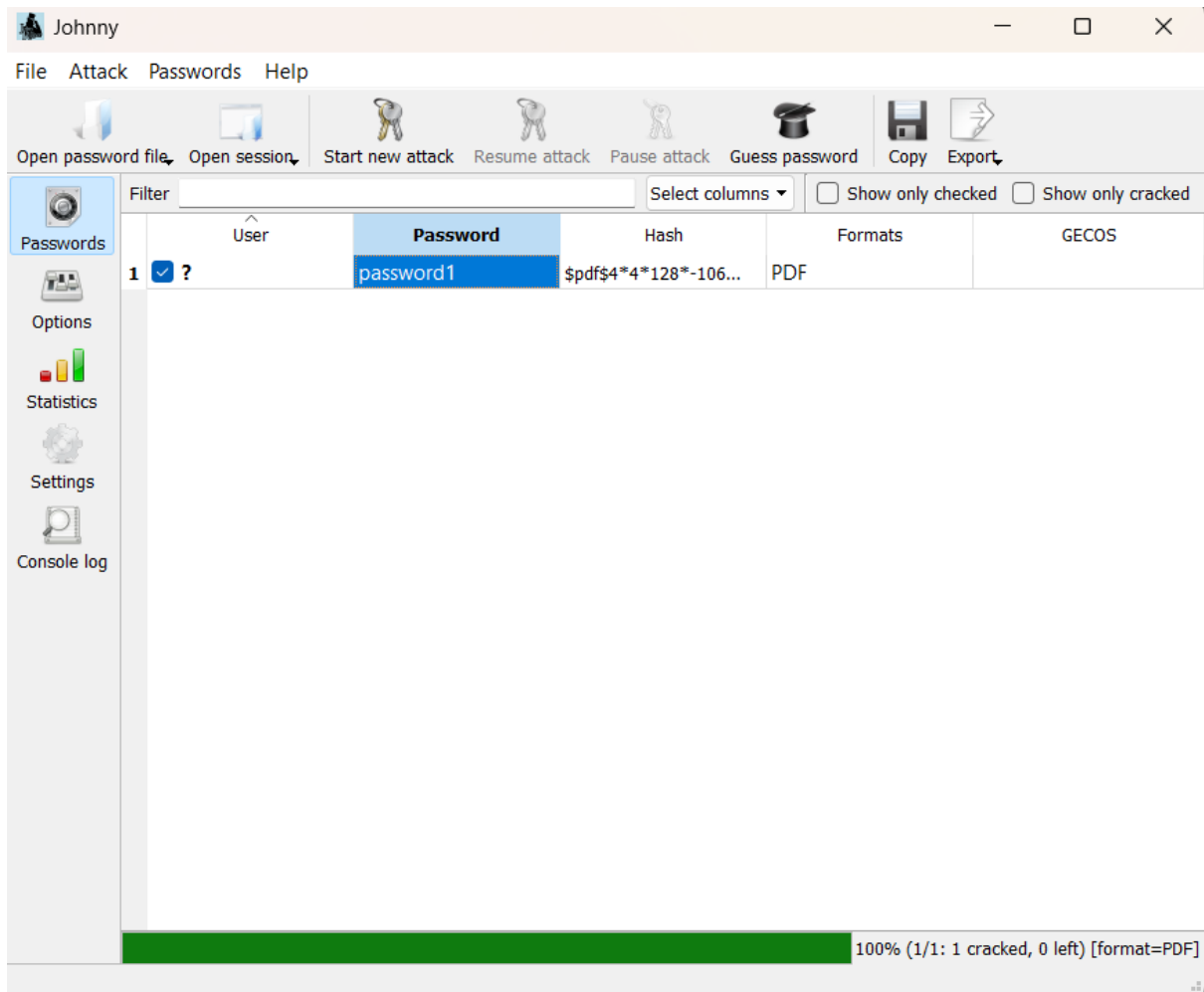
During Week 3 of my Cybersecurity & Ethical Hacking internship, I completed two practical modules focused on password cracking and password security. In the first module, I used John the Ripper (JTR) and Johnny to work with the password hash of the protected PDFs and perform password recovery. In the second module, I used the Networkwalks Hash Calculator to extract the PDFs hash and the Networkwalks Password Cracker to perform the password-cracking process.

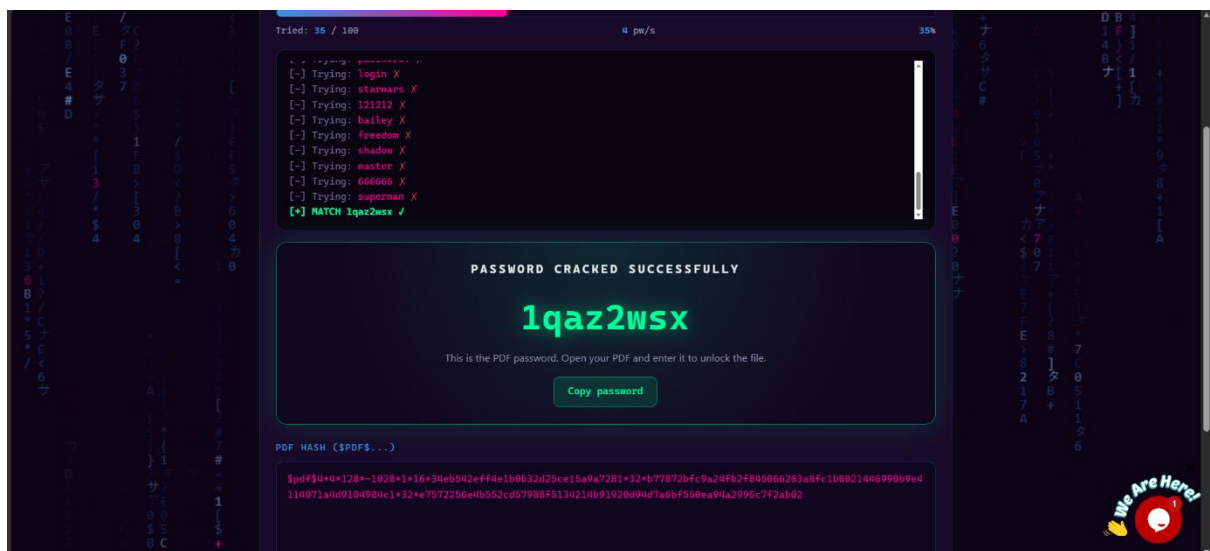
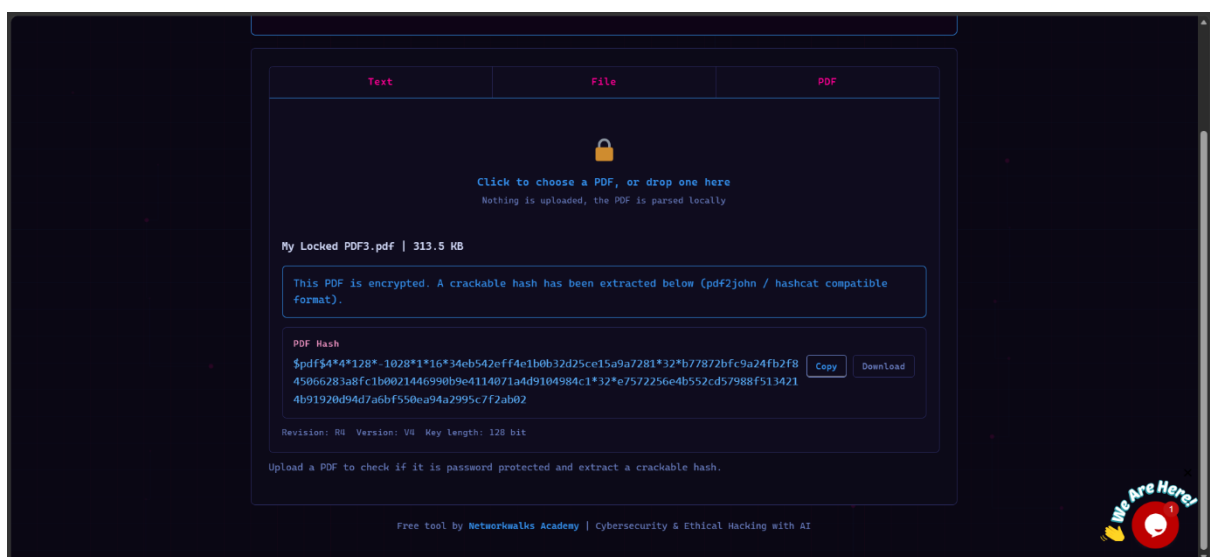
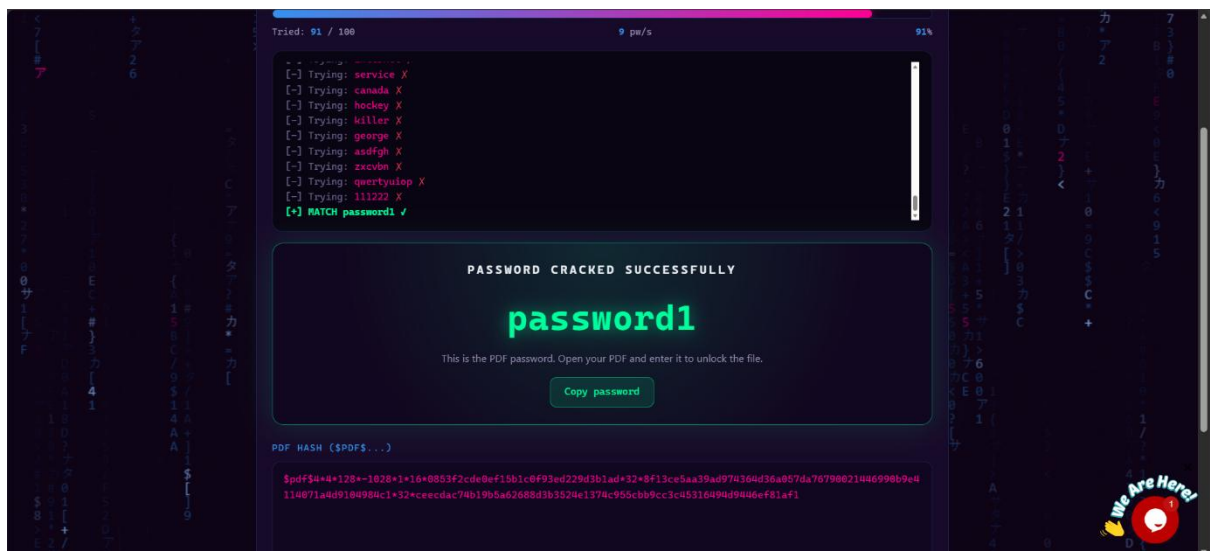
Both activities helped me understand the workflow involved in obtaining a hash, testing password candidates, and verifying the recovered password by opening the protected PDFs. The passwords were successfully recovered during the exercises.

These modules also showed me how easily a common and predictable password can be recovered using password-cracking techniques. Overall, I gained practical experience with JTR, Johnny, password hashes, dictionary-based attacks, and password-security practices. The exercises reinforced the importance of using strong, unique, and unpredictable passwords and performing security testing only within an authorized environment.

## 7. Evidence Collected

[illegible]







## Congratulations!

You have captured your 1<sup>st</sup> flag.

Let's continue this amazing journey of Cyber Security with incredible team of instructors at Networkwalks Academy.

Your Flag1 is: `nw{networkwalks_flag1_jtr_270521_1}`

You can also share this achievement on your Facebook/LinkedIn & tag Networkwalks Academy



## Congratulations!

You have captured another flag.

Cracking passwords is all about patience and the right wordlist.  
You are learning the mindset of a real security tester at Networkwalks.

Your Flag is: `nw{networkwalks_persistence_jtr_270521}`

